

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

Let a triangle PQR be such that P and Q lie on the line $(x+3/8)=(y-4/2)=(z+1/2)=t$ and are at a distance of 6 units from $R(1,2,3)$. If (α, β, γ) is the centroid of ΔPQR , then $\alpha+\beta+\gamma$ is equal to :

- (A) 4
- (B) 5
- (C) 6 **[Correct]**
- (D) 8

Correct Answer: C

Solution Analysis: Write the given line in parametric form. $(x+3/8)=(y-4/2)=(z+1/2)=t$ So any point on the line is $P(t)=(-3+8t, 4+2t, -1+2t)$ Since both P and Q lie on this line and are at a distance 6 from $R(1,2,3)$, we find the points on the line whose distance from R is 6. Distance condition: $\sqrt{(x-1)^2+(y-2)^2+(z-3)^2}=6$ Substitute $x=-3+8t, y=4+2t, z=-1+2t$:
 $(-3+8t-1)^2+(4+2t-2)^2+(-1+2t-3)^2=36$ $(8t-4)^2+(2t+2)^2+(2t-4)^2=36$ Now expand:
 $64t^2-64t+16+4t^2+8t+4+4t^2-16t+16=36$ $72t^2-72t+36=36$ $72t^2-72t=0$ $72t(t-1)=0$ So, $t=0$ or $t=1$